# STRUCTURAL REDUCTIONS AND EXACT RANK CERTIFICATES FOR THE BUFFERED RAMANUJAN ROW MATRICES

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Abstract. This note collects six exact facts used in the finite computations for the buffered Ramanujan row matrices: the parity decomposition, necessary divisibility of a spanning modulus, an exact redundancy among row blocks, the cyclotomic Hall obstruction, the CRT decomposition of the coordinate space, and the sparse rank certificate obtained from block Wiedemann and block Berlekamp–Massey. The purpose is to give one self-contained mathematical account of reductions that otherwise appear only in computational notes. Random choices are used to search for some of the sparse witnesses, but a successful certificate is verified by exact arithmetic over $\mathbb{F}_2$.

## Contents

1. The buffered row system 1
2. The even–odd decomposition 2
3. Necessary divisibility of a spanning modulus 3
4. Exact redundancy of proper multiple blocks 5
5. The cyclotomic Hall constraint 7
6. CRT decomposition of the coordinate space 11
7. A rigorous sparse rank certificate 12
7.1. Sparse square completion 12
7.2. The Berlekamp–Massey lower bound 13
7.3. The matching upper bound 13
7.4. Restoring the full row family 14
8. Implementation map 14

## 1. The buffered row system

Fix positive integers $N$ and $Q$. Let $V_K = K^{\mathbb{Z}/Q\mathbb{Z}}$ for a field $K$, with standard basis $\{e_j : j \in \mathbb{Z}/Q\mathbb{Z}\}$. For $q \mid Q$ and $u \in \mathbb{Z}$ put

$$R_{q,u} = \sum_{\substack{a \bmod q \\ (a,q)=1}} e_{u+aQ/q \bmod Q}. \tag{1}$$

Define

$$A_q = \frac{Q}{\sqrt{\operatorname{lcm}(N, q^2)}}, \qquad \ell_q = \lceil A_q \rceil - 1. \tag{2}$$

The strict interval occurring in the definition of the buffered family is exactly

$$I_{N,Q}(q) = \mathbb{Z} \cap (-A_q, A_q - 1) = \{-\ell_q, \dots, \ell_q - 1\}. \tag{3}$$

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The last set is empty when $\ell_q = 0$. We write

$$\mathcal{R}_N(Q) = \{R_{q,u} : q \mid Q,\ u \in I_{N,Q}(q)\}. \tag{4}$$

All rows have integral entries, so they can be considered over $\mathbb{Q}$, $\mathbb{C}$, or a finite field. Full rank over $\mathbb{F}_p$ implies full rank over $\mathbb{Q}$: a nonzero maximal minor modulo $p$ is a nonzero integral minor.

*Status.* All identities and rank bounds below are proved exactly. The small-range checks mentioned in the text are independent tests of the formulae, not substitutes for the proofs. In the block-Wiedemann section, a run that does not satisfy the displayed certificate equalities is explicitly *inconclusive*.

## 2. The even–odd decomposition

Assume throughout this section that $Q$ is even, and put

$$V_b = \operatorname{span}_K\{e_j : j \equiv b \pmod 2\} \qquad (b = 0, 1).$$

**Proposition 2.1** (Parity support and reflection)**.** *Every row $R_{q,u}$ is contained in exactly one of $V_0, V_1$. Moreover the coordinate reflection*

$$\tau e_j = e_{Q-1-j} \tag{5}$$

*satisfies*

$$\tau R_{q,u} = R_{q,-u-1}. \tag{6}$$

*The involution $u \mapsto -u-1$ preserves $I_{N,Q}(q)$. Consequently, for any union of full buffered $q$-blocks, its row span $S$ decomposes as*

$$S = (S \cap V_0) \oplus (S \cap V_1), \qquad \tau(S \cap V_0) = S \cap V_1. \tag{7}$$

*In particular the two sector ranks and the two sector nullities are equal.*

*Proof.* Write $h = Q/q$. If $h$ is even, every support index $u + ah$ in (1) has parity $u$. If $h$ is odd, then $q$ is even, and every unit $a \bmod q$ is odd. Every support index then has parity $u + 1$. This proves the first assertion.

Replacing $a$ by $-a \bmod q$ gives

$$Q - 1 - (u + aQ/q) \equiv -u - 1 + (-a)Q/q \pmod Q,$$

which proves (6). From (3), the two endpoints $-\ell_q$ and $\ell_q - 1$ are interchanged by $u \mapsto -u - 1$. Finally $Q - 1$ is odd, so $j \mapsto Q - 1 - j$ interchanges the coordinate parities. This proves (7) and the rank assertions. □

**Corollary 2.2** (One parity suffices)**.** *For the full buffered family at even $Q$, full rank in either coordinate parity implies full rank in both. In particular the full defect is even. A dense computation on one sector uses one quarter of the quadratic storage and one eighth of the cubic arithmetic required by the unsplit matrix.*

The same decomposition is compatible with the top-up used at prime level. Suppose $Q = NM$, with $M$ even, and define

$$\omega_c = \sum_{k \bmod N} e_{c+kM}, \qquad W_N = \operatorname{span}_K\{\omega_c : c \bmod M\}. \tag{8}$$

Every $\omega_c$ has coordinate parity $c$, and

$$\tau\omega_c = \omega_{-c-1 \bmod M}. \tag{9}$$

Thus $W_N$ has two isomorphic parity summands of dimension $M/2$. Orthogonal projection away from $W_N$, the assisted kernel, and the orbit-sum map on the coprime-row kernel all split into two

isomorphic blocks. If the final defect in one block is $\delta$, the full defect is $2\delta$. In particular a projected rank lower bound $Q - M - 1$ already forces the full projected rank $Q - M$.

## 3. Necessary divisibility of a spanning modulus

We first record a dilation observation.

**Lemma 3.1** (Dilation)**.** *If $\mathcal{R}_N(Q)$ spans $K^Q$, then $\mathcal{R}_N(tQ)$ spans $K^{tQ}$ for every positive integer $t$.*

*Proof.* For $0 \le c < t$ define

$$\iota_c(e_j) = e_{tj+c} \colon K^Q \longrightarrow K^{tQ}.$$

The $t$ images have disjoint support and their direct sum is $K^{tQ}$. A direct calculation gives

$$\iota_c(R^Q_{r,u}) = R^{tQ}_{r,tu+c}. \tag{10}$$

If $-A_r < u < A_r - 1$, then

$$-tA_r < tu + c < t(A_r - 1) + (t - 1) = tA_r - 1.$$

The radius at $tQ$ is $tA_r$, so the row on the right of (10) is legal. Each $\iota_c(K^Q)$ is therefore contained in the new row span. □

**Theorem 3.2** (Forced factors)**.** *Suppose that $\mathcal{R}_N(Q)$ spans. Then*

1. *every prime $p \ge 11$ dividing $N$ also divides $Q$;*
2. *if $4 \mid N$, then $2 \mid Q$.*

*Equivalently,*

$$2^{\mathbf{1}_{4|N}} \prod_{\substack{p \mid N \\ p \ge 11}} p \;\mid Q. \tag{11}$$

*Proof.* For the first assertion, suppose that $p \ge 11$, $p \mid N$, and $p \nmid Q$. By Lemma 3.1, the family still spans after replacing $Q$ by $Q' = \operatorname{lcm}(Q, 3)$. The prime $p$ does not divide $Q'$. Consider

$$v = e_{Q'/3} - e_{2Q'/3}. \tag{12}$$

For every $r \mid Q'$ we have

$$\operatorname{lcm}(N, r^2) \ge pr^2.$$

Every legal shift therefore satisfies

$$\frac{|u|}{Q'} < \frac{1}{r\sqrt{p}} < \frac{1}{3r}. \tag{13}$$

If $R_{r,u}$ meets $kQ'/3$, where $k = 1$ or $2$, then for some reduced $a/r$,

$$\frac{a}{r} + \frac{u}{Q'} \equiv \frac{k}{3} \pmod 1.$$

Two unequal reduced fractions of denominators $r$ and $3$ have circular distance at least $1/\operatorname{lcm}(r, 3) \ge 1/(3r)$. Hence (13) forces $r = 3$, $a = k$, and $u = 0$. The only row that can meet either coordinate of (12) therefore meets both, with total pairing $1 - 1 = 0$. Thus $v$ annihilates the full row family, a contradiction.

For the second assertion, suppose that $4 \mid N$ and $Q$ is odd. Every $r \mid Q$ is odd. Put $j = (Q-1)/2$. Since $\sqrt{\operatorname{lcm}(N, r^2)} \ge 2r$, every legal shift satisfies

$$\frac{|u + 1/2|}{Q} < \frac{A_r - 1/2}{Q} < \frac{1}{2r}. \tag{14}$$

If a row met $j$, then for some reduced $a/r$,

$$\frac{a}{r} - \frac{1}{2} \equiv -\frac{u + 1/2}{Q} \pmod 1.$$

Because $r$ is odd, the circular distance from $a/r$ to $1/2$ is at least $1/(2r)$, contradicting (14). Thus every row misses the coordinate $j$, so spanning is impossible. □

**Theorem 3.3** (Sharpness of the level hypothesis)**.** *Suppose that $s^2 \mid N$ and $\varphi(s) > 1$. Then $\mathcal{R}_N(Q)$ fails to span $K^{\mathbb{Z}/Q\mathbb{Z}}$ for every positive integer $Q$ and every field $K$.*

*More precisely, put $Q' = \operatorname{lcm}(Q, s)$. The orthogonal complement of $\operatorname{span}_K \mathcal{R}_N(Q')$ contains the $(\varphi(s) - 1)$-dimensional space*

$$\left\{ \sum_{\substack{a \bmod s \\ (a,s)=1}} c_a e_{aQ'/s} : \sum_{\substack{a \bmod s \\ (a,s)=1}} c_a = 0 \right\}. \tag{15}$$

*Consequently, the existence of a spanning modulus $Q$ forces*

$$16 \nmid N \qquad \text{and} \qquad p^2 \nmid N \quad \text{for every odd prime } p. \tag{16}$$

*Conversely, the failure of* (16) *always supplies such an $s$. Thus the hypotheses on $N$ in the spanning conjecture cannot be relaxed.*

*Proof.* Suppose first that $Q'$ is a multiple of $s$, and let $v$ be an element of (15). Fix a legal row $R_{r,u}$, where $r \mid Q'$. Since $s^2 \mid N$, one has

$$\operatorname{lcm}(s, r)^2 = \operatorname{lcm}(s^2, r^2) \mid \operatorname{lcm}(N, r^2). \tag{17}$$

The strict buffered window therefore gives

$$\frac{|u|}{Q'} < \frac{1}{\sqrt{\operatorname{lcm}(N, r^2)}} \le \frac{1}{\operatorname{lcm}(s, r)}. \tag{18}$$

Suppose that $R_{r,u}$ meets one of the coordinates $aQ'/s$ occurring in $v$. Then for some reduced residue $b \bmod r$,

$$\frac{b}{r} + \frac{u}{Q'} \equiv \frac{a}{s} \pmod 1. \tag{19}$$

Two distinct reduced fractions of denominators $r$ and $s$ have circular distance at least $1/\operatorname{lcm}(r, s)$. Equations (18) and (19) therefore force $b/r = a/s$ in $\mathbb{R}/\mathbb{Z}$. Uniqueness of reduced denominators gives $r = s$ and $b = a$. Equation (19) then makes $u/Q'$ integral, while (18) forces $u = 0$.

It follows that every legal row misses the support of $v$, except possibly $R_{s,0}$. That exceptional row contains every primitive $s$-torsion coordinate once, and hence

$$\langle R_{s,0}, v \rangle = \sum_{(a,s)=1} c_a = 0.$$

Thus the whole space (15) annihilates $\mathcal{R}_N(Q')$. The coordinate vectors in its support are distinct, and the single coefficient-sum equation has rank one, so this space has dimension $\varphi(s) - 1 > 0$.

Now let $Q$ be arbitrary. If $\mathcal{R}_N(Q)$ spanned, the dilation lemma would imply spanning at $Q' = \operatorname{lcm}(Q, s)$, contradicting the packet just constructed. This proves the first assertion.

If an odd prime square $p^2$ divides $N$, take $s = p$. If $16 \mid N$, take $s = 4$. In both cases $s^2 \mid N$ and $\varphi(s) > 1$. Conversely, suppose $s^2 \mid N$ and $\varphi(s) > 1$. If an odd prime $p$ divides $s$, then $p^2 \mid N$. Otherwise $s$ is a power of two; the totient condition forces $4 \mid s$, and hence $16 \mid N$. This proves both the equivalence and the sharpness assertion. □

For a level satisfying (16), Theorem 3.2 therefore supplies the full possible valuation at every prime $p \geq 11$. There is no analogous forced factor for $3, 5$, or $7$ at exponent one: the spanning examples $(N, Q) = (3, 2), (5, 6), (7, 6)$ omit the corresponding prime. Likewise $4 \mid N$ forces only $2 \mid Q$; the examples $(N, Q) = (4, 6)$ and $(8, 6)$ show that $4 \mid Q$ is not forced. These examples are elementary: at $Q = 6$, the legal $q = 1$ rows give $e_4, e_5, e_0, e_1$, while the legal $q = 2$ rows with shifts $-1, 0$ give $e_2, e_3$; the same six rows are legal at levels $4, 5, 7$, and $8$. At $(N, Q) = (3, 2)$ the two legal $q = 1$ rows are already the two coordinate vectors.

## 4. Exact redundancy of proper multiple blocks

For $q \mid Q$ define the complete row

$$C_{q,u} = \sum_{a \bmod q} e_{u+aQ/q}. \tag{20}$$

Partitioning the residues modulo $q$ according to the denominator of the corresponding reduced fraction gives the fundamental identity

$$C_{q,u} = \sum_{d \mid q} R_{d,u}, \qquad R_{q,u} = \sum_{d \mid q} \mu(q/d) C_{d,u}. \tag{21}$$

If $d \mid q$, then

$$\operatorname{lcm}(N, d^2) \mid \operatorname{lcm}(N, q^2), \tag{22}$$

so every shift legal at $q$ is also legal at $d$.

**Proposition 4.1** (Orbit quotient bound)**.** *Fix $s \mid Q$, and let*

$$W_s = \operatorname{span}_K \{C_{s,v} : v \in \mathbb{Z}\}. \tag{23}$$

*Then $\dim W_s = Q/s$, and every legal primitive row with $s \mid q$ belongs to the span of $W_s$ and the legal blocks whose moduli are not divisible by $s$. Consequently*

$$\operatorname{rank} \mathcal{R}_N(Q) \leq \frac{Q}{s} + \sum_{\substack{q \mid Q \\ s \nmid q}} \min\left(2\ell_q, \frac{Q}{q}\right). \tag{24}$$

*Taking the minimum over $s \mid Q$ gives a rigorous prefilter for candidate moduli.*

*Proof.* The $Q/s$ vectors $C_{s,v}$ with $v$ in a complete residue system modulo $Q/s$ have disjoint supports, so they form a basis of $W_s$. If $q = sm$, then

$$C_{q,u} = \sum_{b=0}^{m-1} C_{s,u+bQ/q} \in W_s. \tag{25}$$

Now use the first identity in (21) and induct over $q/s$. It expresses $R_{q,u}$ as $C_{q,u}$ minus rows at proper divisor moduli. Terms still divisible by $s$ are handled by the induction, and all other terms belong to the stated blocks. Their shifts remain legal by (22).

For completeness, order the blocks with $s \nmid q$ compatibly with divisibility. Modulo blocks at proper divisors of $q$, the second identity in (21) identifies $R_{q,u}$ with $C_{q,u}$. It is the complete row $C_{q,u}$, not the primitive row $R_{q,u}$, which is periodic in $u$ with period $Q/q$. Thus the contribution of the $q$-block modulo the earlier blocks is at most $\min(2\ell_q, Q/q)$. Summing these quotient bounds and adding $\dim W_s$ proves (24). □

**Theorem 4.2** (Deletion at a saturated block)**.** *Suppose that $s \mid Q$ and*

$$2\ell_s \geq \frac{Q}{s}. \tag{26}$$

*Then*

$$\operatorname{span}_K \mathcal{R}_N(Q) = \operatorname{span}_K\{R_{q,u} \in \mathcal{R}_N(Q) : q = s \text{ or } s \nmid q\}. \tag{27}$$

*Thus every block with $s \mid q$ and $q \neq s$ may be omitted without changing the row span.*

*Proof.* The legal interval at $s$ consists of $2\ell_s$ consecutive integers. By (26), it meets every residue class modulo $Q/s$. Consequently the legal complete $s$-rows span all of $W_s$. Each such complete row is, by (21), a sum of the legal $q = s$ row and rows at proper divisors of $s$; those proper divisors are not divisible by $s$. Proposition 4.1 now gives (27). □

The previous $N$-only criterion is the following useful special case. Put

$$s_N = \prod_{p^a \| N} p^{\lceil a/2 \rceil}, \tag{28}$$

the least positive integer satisfying $N \mid s_N^2$. If $s_N \mid Q$ and $Q/s_N \geq 2$, then

$$\operatorname{lcm}(N, s_N^2) = s_N^2, \qquad \ell_{s_N} = Q/s_N - 1,$$

so (26) holds. When $Q = s_N$, there is no proper $s_N$-multiple divisor of $Q$, and the deletion assertion is vacuous.

This strictly generalizes the prime-level rule: for prime $N \mid Q$, retain $q = N$ and omit $2N, 3N, \ldots$. For a nonsquarefree admissible level it can be stronger; for example $s_{92} = 46$.

**Theorem 4.3** (Simultaneous triangular pruning)**.** *Define*

$$\mathcal{A} = \{s \mid Q : 2\ell_s \geq Q/s\}$$

*and let $\mathcal{A}_{\min}$ be its divisibility-minimal elements. Delete a modulus $q \mid Q$ if it is a proper multiple of some $s \in \mathcal{A}_{\min}$. The rows in all deleted blocks lie in the span of the retained blocks.*

*Moreover, for each $s \in \mathcal{A}_{\min}$ choose a subset $U_s \subseteq I_{N,Q}(s)$ containing one representative of every residue class modulo $Q/s$. The full row span is unchanged if the retained $s$-block is replaced by $\{R_{s,u} : u \in U_s\}$. Thus there is an exact spanning subfamily with*

$$\sum_{\substack{q \mid Q \\ q \text{ not a proper multiple of any } s \in \mathcal{A}_{\min}}} \min\left(2\ell_q, \frac{Q}{q}\right) \tag{29}$$

*rows and the same span as $\mathcal{R}_N(Q)$.*

*Proof.* We use induction in the divisor order. Suppose first that $q$ is deleted and choose $s \in \mathcal{A}_{\min}$ properly dividing $q$. By (25), $C_{q,u} \in W_s$. The proof of Theorem 4.2 expresses every element of $W_s$ using legal primitive rows at divisors of $s$, hence at proper divisors of $q$. On the other hand,

$$R_{q,u} = C_{q,u} - \sum_{\substack{d \mid q \\ d < q}} R_{d,u}.$$

All displayed rows are legal by (22). Every deleted proper-divisor block in this expression has already been reduced by the induction, proving the simultaneous deletion statement. This strict descent also shows that combining different values of $s$ introduces no cycle.

Now let $s \in \mathcal{A}_{\min}$ and $u \in I_{N,Q}(s)$. Choose $v \in U_s$ congruent to $u$ modulo $Q/s$. Since $C_{s,u} = C_{s,v}$, Möbius inversion gives

$$R_{s,u} - R_{s,v} = \sum_{\substack{d \mid s \\ d < s}} (R_{d,v} - R_{d,u}).$$

Again every row on the right is legal and belongs to a proper-divisor block. Induction proves the shift reduction. A nonsaturated retained block contributes all $2\ell_q$ rows, while a saturated retained block is an element of $\mathcal{A}_{\min}$ and contributes $Q/q$ representatives. This gives (29). □

If $Q/s$ is even, the representatives may be chosen compatibly with the reflection used in Section 2:

$$U_s = \{u, -u - 1 : 0 \le u < Q/(2s)\}.$$

Thus the shift reduction retains exactly $Q/(2s)$ rows in each coordinate parity. When $Q/s$ is odd, the full-family reduction remains valid, but a one-parity implementation should either retain the whole anchor block or handle the two parities together.

Finally, the stronger condition is still quite restrictive. Put $r = N/\gcd(N, s^2)$ and $h = Q/s$. Then $\ell_s = \lceil h/\sqrt{r} \rceil - 1$. If $r \ge 4$, then

$$2\ell_s \le 2\lceil h/2 \rceil - 2 < h,$$

so every saturated $s$ satisfies

$$\frac{N}{\gcd(N, s^2)} \le 3. \tag{30}$$

*Exact checks.* Direct elimination over $\mathbb{F}_2$ verified (27) for all 1605 triples

$$1 \le N \le 18, \qquad 1 \le Q \le 96, \qquad s \mid Q, \qquad N \mid s^2.$$

The bound (24) was independently checked in all 8262 corresponding single-divisor cases. Old and new implementations at $(N, Q) = (12, 60)$ and $(20, 60)$ returned identical ranks, while the new FLINT and composite-CRT constructors processed fewer rows.

The simultaneous implementation was also compared directly with the preceding single-$s_N$ implementation at levels where the new criterion is strictly stronger. These comparisons include both spanning and nonspanning matrices:

| $N$ | $Q$ | minimal saturated anchors | common full defect |
|---|---|---|---|
| 38 | $798, 1064, 1330, 1596$ | 19 | $88, 148, 258, 0$ |
| 66 | $660, 1320, 1980, 2640, 3300$ | $22, 33$ | $8, 12, 20, 24, 0$ |
| 78 | $780, 1560, 2340, 5460$ | $26, 39$ | $50, 58, 138, 0$ |

Here $N = 38$ was checked over $\mathbb{Q}$ with FLINT, $N = 66$ by direct elimination over $\mathbb{F}_2$ (with independent compressed-RRE and $\mathbb{F}_{127}$ checks at the spanning endpoint), and $N = 78$ with both serial and parallel composite-CRT finishers. At $(38, 1596)$ the FLINT residual matrix shrank from $496 \times 357$ to $409 \times 357$. At $(78, 5460)$ the retained full-row count fell from 6854 to 6396. All comparisons use exact arithmetic.

## 5. The cyclotomic Hall constraint

The complete rows also give a fast necessary condition for spanning. For each retained $q \mid Q$ put

$$b_q = \min\left(2\ell_q, \frac{Q}{q}\right). \tag{31}$$

For the sharpest version, apply Theorem 4.3 first: "retained" means that $q$ is not a proper multiple of any divisibility-minimal saturated anchor. Equivalently, one may leave every divisor in the graph and assign capacity zero to the deleted blocks.

Make a capacitated bipartite graph. Its left vertices are the retained divisors $q \mid Q$, with capacity $b_q$. Its right vertices are all divisors $D \mid Q$, with capacity $\varphi(D)$. Put an edge

$$q \longrightarrow D \quad \Longleftrightarrow \quad D \mid Q/q. \tag{32}$$

Let $F_N(Q)$ denote the maximum flow from the left capacities to the right capacities.

**Theorem 5.1** (Cyclotomic Hall obstruction)**.** *Over $\mathbb{Q}$ one has*

$$\operatorname{rank} \mathcal{R}_N(Q) \le F_N(Q), \qquad \operatorname{nullity} \mathcal{R}_N(Q) \ge Q - F_N(Q). \tag{33}$$

*In particular spanning requires $F_N(Q) = Q$.*

*Proof.* Identify the ambient space with $\mathbb{Q}[X]/(X^Q - 1)$, and write

$$P_q(X) = \sum_{(a,q)=1} X^{aQ/q}, \qquad C_q(X) = \sum_{a \bmod q} X^{aQ/q}.$$

The identities (21) give an invertible divisor-triangular change from legal primitive rows to legal complete rows. The nesting (22) is what makes this change legal at the exact buffered endpoints.

The $q$-block of complete rows has rank exactly $b_q$: it consists of consecutive members of a set of $Q/q$ independent, disjoint-support translates. We may therefore replace it by any basis of $b_q$ of its rows without changing the full row span. If $\zeta_D$ is a primitive $D$th root of unity, then

$$C_q(\zeta_D) = \begin{cases} q, & D \mid Q/q, \\ 0, & D \nmid Q/q. \end{cases} \tag{34}$$

Indeed, this is the geometric sum

$$\sum_{a \bmod q} \zeta_D^{aQ/q};$$

it is $q$ when $\zeta_D^{Q/q} = 1$, equivalently when $D \mid Q/q$, and is zero otherwise.

The Chinese remainder theorem gives a $\mathbb{Q}$-linear isomorphism

$$\mathbb{Q}[X]/(X^Q - 1) \longrightarrow \bigoplus_{D \mid Q} \mathbb{Q}[X]/(\Phi_D) \cong \bigoplus_{D \mid Q} \mathbb{Q}(\zeta_D), \qquad f \longmapsto \big(f(\zeta_D)\big)_{D \mid Q}. \tag{35}$$

Under this map the row $X^u C_q(X)$ has $D$-component

$$\zeta_D^u C_q(\zeta_D). \tag{36}$$

Choose a $\mathbb{Q}$-basis of each field $\mathbb{Q}(\zeta_D)$. The transformed ordinary matrix now has row blocks indexed by $q$, containing $b_q$ rows, and column blocks indexed by $D$, containing $\varphi(D)$ columns. Equations (34) and (36) say that its $(q, D)$ block is zero unless $D \mid Q/q$. Both the row-basis replacements and (35) preserve rank.

We spell out the remaining linear-algebra bound. Replace each left vertex $q$ by $b_q$ formal copies and each right vertex $D$ by $\varphi(D)$ formal copies, joining every copy of $q$ to every copy of $D$ whenever $D \mid Q/q$. The *capacitated block term rank* is the largest matching in this expanded bipartite graph. Equivalently, it is the largest number of positions one could select in the permitted blocks with no two positions in the same ordinary row or ordinary column. Collapsing the copies turns such a matching into an integral flow with source capacity $b_q$ at $q$ and sink capacity $\varphi(D)$ at $D$. Conversely, an integral flow can be assigned to distinct unused copies at its two endpoints and so gives such a matching. Since all capacities are integral, the integral max-flow theorem shows that this term rank is exactly $F_N(Q)$.

Finally, let the actual transformed matrix have rank $r$. It has a nonzero $r \times r$ minor. In the determinant expansion of that minor, at least one permutation term is nonzero. The $r$ entries

in that term lie in distinct ordinary rows and columns, and a nonzero entry can occur only in a permitted $(q, D)$ block. They therefore give a matching of size $r$ in the expanded graph. Hence

$$r \le F_N(Q).$$

Notice that the reverse inequality need not hold: the expanded graph treats every permitted block as potentially nonzero and ignores both internal zeros and cancellation in its minors. This is precisely why the Hall test is a necessary condition rather than a rank formula. The elementary matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ already illustrates the distinction: its permitted-entry pattern has a matching of size two, but cancellation makes its matrix rank one. Since the ambient space has dimension $Q$, the asserted nullity bound follows as well. □

Equivalently, every set $\mathcal{T}$ of right-hand divisors must satisfy the Hall inequality

$$\sum_{D\in\mathcal{T}} \varphi(D) \le \sum_{\substack{q \text{ retained} \\ D\mid Q/q \text{ for some } D\in\mathcal{T}}} b_q. \tag{37}$$

A useful cheap family of cuts is obtained as follows. Choose a nonempty set $\mathcal{S}$ of primes dividing $Q$, and let

$$B_{\mathcal{S}} = \prod_{p\in\mathcal{S}} p^{v_p(Q)}.$$

The target components divisible by $B_{\mathcal{S}}$ have total dimension

$$\sum_{\substack{D\mid Q \\ B_{\mathcal{S}}\mid D}} \varphi(D) = \varphi(B_{\mathcal{S}})\frac{Q}{B_{\mathcal{S}}},$$

and only $q$ coprime to $B_{\mathcal{S}}$ can reach them. Hence

$$\operatorname{rank}\mathcal{R}_N(Q) < Q \quad \text{if} \quad \sum_{\substack{q\mid Q \\ (q,B_{\mathcal{S}})=1}} b_q < \varphi(B_{\mathcal{S}})\frac{Q}{B_{\mathcal{S}}}. \tag{38}$$

**Theorem 5.2** (Existence of Hall-admissible moduli)**.** *For every positive integer $N$ there are arbitrarily large positive integers $Q$, divisible by any prescribed positive integer, such that*

$$F_N(Q) = Q. \tag{39}$$

*More precisely, for any $N \ge 4$ and any $Q \ge 1$, the sufficient condition*

$$\frac{2}{\sqrt{N}}\sigma_1(Q) - 2\tau(Q) \ge Q, \qquad \sigma_1(Q) = \sum_{e\mid Q} e, \quad \tau(Q) = \sum_{e\mid Q} 1, \tag{40}$$

*implies* (39)*. No coprimality, square-freeness, or divisibility relation between $N$ and $Q$ is required.*

*Proof.* First suppose $N \ge 4$ and work with the graph before pruning. Put $a = 2/\sqrt{N} \le 1$ and index a source by $e = Q/q$. Since $\operatorname{lcm}(N, q^2) \mid Nq^2$, its capacity satisfies

$$b_{Q/e} = \min\{2\ell_{Q/e}, e\} \ge ae - 2. \tag{41}$$

Indeed, $A_{Q/e} \ge e/\sqrt{N}$ and $\lceil A_{Q/e}\rceil - 1 \ge A_{Q/e} - 1$; also $e \ge ae - 2$. The possibly negative lower bound for small $e$ causes no difficulty.

The source $e$ reaches a target $D$ exactly when $D \mid e$. On the divisor lattice of $Q$, consider the three probability measures

$$\mu_1(e) = \frac{e}{\sigma_1(Q)}, \qquad \mu_0(e) = \frac{1}{\tau(Q)}, \qquad \mu_2(e) = \frac{\varphi(e)}{Q}. \tag{42}$$

Each is a product measure on the prime exponents. At a prime power $p^A \| Q$, the probabilities of an exponent $j$ are respectively proportional to $p^j$, 1, and $\varphi(p^j)$. For $1 \le k \le A$, their cumulative probabilities satisfy

$$\frac{p^k - 1}{p^{A+1} - 1} \le p^{k-1-A}, \qquad \frac{p^k - 1}{p^{A+1} - 1} \le \frac{k}{A+1}. \tag{43}$$

The first inequality is equivalent to $p^A \ge p^{k-1}$; the second follows because the weights $1, p, \dots, p^A$ increase. Coupling the exponents by the same uniform random number at each prime makes the $\mu_1$ divisor a multiple of the $\mu_0$ and $\mu_2$ divisors. Thus for every upward-closed set $\mathcal{U}$,

$$\mu_1(\mathcal{U}) \ge \mu_0(\mathcal{U}), \qquad \mu_1(\mathcal{U}) \ge \mu_2(\mathcal{U}).$$

Using (41) and (40), we obtain

$$\begin{aligned} \sum_{e \in \mathcal{U}} b_{Q/e} &\ge a\sigma_1(Q)\mu_1(\mathcal{U}) - 2\tau(Q)\mu_0(\mathcal{U}) \\ &\ge \big(a\sigma_1(Q) - 2\tau(Q)\big)\mu_1(\mathcal{U}) \\ &\ge Q\mu_2(\mathcal{U}) = \sum_{e \in \mathcal{U}} \varphi(e). \end{aligned} \tag{44}$$

For an arbitrary target set $\mathcal{T}$, its source neighbourhood is the upward closure $\mathcal{U}$ of $\mathcal{T}$. Since $\mathcal{T} \subseteq \mathcal{U}$, this gives

$$\sum_{e \in \mathcal{U}} b_{Q/e} \ge \sum_{D \in \mathcal{T}} \varphi(D).$$

Hence there is a flow of value $Q$ before pruning.

Pruning does not destroy this flow. A divisibility-minimal saturated source $s$ is retained and has capacity $Q/s$, exactly the total target capacity in its neighbourhood:

$$\sum_{D \mid Q/s} \varphi(D) = Q/s.$$

Assign each deleted source $q$ to one such $s \mid q$, and transfer its outgoing flow to $s$. Every target reached by $q$ is also reached by $s$. The resulting total flow from any $s$, including its old flow, is at most the total capacity of its neighbourhood, namely $Q/s$. Thus all source and target capacities are respected, even when the saturated neighbourhoods overlap. This proves (39) for the retained graph.

Finally, fix any prescribed divisor $B$, and take $Q = BR$ with $R$ square-free on primes not dividing $B$. Multiplicativity gives

$$\frac{\sigma_1(Q)}{Q} = \frac{\sigma_1(B)}{B} \prod_{p \mid R} \left(1 + \frac{1}{p}\right), \qquad \frac{\tau(Q)}{Q} = \frac{\tau(B)}{B} \prod_{p \mid R} \frac{2}{p}.$$

As $R$ includes successively more primes, the first product diverges and the second is bounded. For example, the divergence follows from that of $\prod_p (1-1/p)^{-1}$ and the identity $1+1/p = (1-1/p^2)/(1-1/p)$. Consequently (40) eventually holds. For $N \le 3$, the same conclusion follows by using the construction for $N = 4$: the bound (41) with $a = 1$ remains valid, since $A_{Q/e} \ge e/\sqrt{N} \ge e/2$. □

*Remark* 5.3 (What the Hall condition does not see). Theorem 5.2 does not use the square-free-level hypothesis (16). Thus the Hall obstruction does not detect the torsion packets of Theorem 3.3. It also does not imply either forced factor in Theorem 3.2. For example, direct flows give

$$F_{11}(30) = 30 \quad (11 \nmid 30), \qquad F_4(15) = 15 \quad (2 \nmid 15). \tag{45}$$

For the first equality one may send capacities

$$8 : q = 2 \to D = 15, \quad 4 : q = 3 \to D = 10, \quad 2 : q = 5 \to D = 6, \quad 2 : q = 6 \to D = 5,$$

and use $q = 1$ for capacities $8, 2, 2, 1, 1$ at $D = 30, 5, 3, 2, 1$. For the second, send capacities $4 : q = 3 \to D = 5$ and $2 : q = 5 \to D = 3$, and use $q = 1$ for $8 : D = 15$ and $1 : D = 1$.

The reason is structural. The Hall graph remembers only which cyclotomic blocks can be nonzero and their dimensions. It replaces every permitted block by a generic block and forgets the actual vectors $\zeta_D^u$, the fact that the same shift $u$ controls all its components, and cancellations between those components. The square-factor packets and the two forced-factor obstructions all live in this discarded internal structure. Thus Theorem 5.2 removes a purely dimensional obstruction and supplies evidence for the spanning conjecture at admissible levels, but the missing step from Hall admissibility to spanning must use the arithmetic of the nonzero blocks (equivalently, the spatial Farey geometry), not merely a finer analysis of the same zero pattern.

*Remark* 5.4. The Hall condition is necessary, not sufficient. It records the cyclotomic zero pattern but ignores arithmetic dependence among the nonzero blocks. A candidate with $F_N(Q) = Q$ must still undergo an exact rank calculation.

*Remark* 5.5 (Maximum flow does not by itself select spanning rows). A maximum flow can be made deterministic and used to assign quotas to the $q$-blocks whose sum is $F_N(Q)$. Choosing that many canonical shifts from each block gives a useful candidate row set, but the Hall argument does not prove that it spans the original rows. The flow uses only the zero pattern (32); it does not see cancellations among the nonzero cyclotomic blocks. In particular, when $F_N(Q) = Q$, proving that a flow-selected set of $Q$ rows is independent is exactly an additional nonvanishing-minor statement, not a consequence of Hall's theorem.

There is of course an a posteriori canonical construction: order all rows, perform exact elimination, retain the first row that increases rank, and, if desired, pad the resulting basis to $F_N(Q)$ rows. This preserves the row space, but it has already performed the rank computation that the Hall screen was meant to precede. Thus a genuinely combinatorial canonical $F_N(Q)$-row reduction would require a new arithmetic theorem about these particular nonzero blocks.

## 6. CRT decomposition of the coordinate space

Let $Q = PL$, where $P$ is a prime factor chosen for the split. Define

$$V_r = \operatorname{span}_K\{e_{r+Pt} : t \bmod L\}, \qquad r \bmod P. \tag{46}$$

Then

$$V_K = \bigoplus_{r \bmod P} V_r, \qquad V_r \cong K^{\mathbb{Z}/L\mathbb{Z}}. \tag{47}$$

This is simply the unique representation $j = r + Pt$, with $0 \le r < P$ and $t \bmod L$.

If $q \mid Q$ and $P \nmid q$, then $q \mid L$ and $Q/q = P(L/q)$. Writing $u = r + Pv$, the row becomes, within $V_r$,

$$R_{q,r+Pv} \longleftrightarrow \sum_{(a,q)=1} e^{(r)}_{v+aL/q}. \tag{48}$$

Thus every $P$-free row stays in one sector. If $P$ occurs to exact exponent one in $Q$, the rows with $P \mid q$ generally meet several sectors and provide the coupling between them.

The legality of $u = r + Pv$ gives the exact sector interval

$$\left\lceil \frac{-\ell_q - r}{P} \right\rceil \le v \le \left\lfloor \frac{\ell_q - 1 - r}{P} \right\rfloor. \tag{49}$$

For nearby residues $r$ these intervals have a large common core and differ only in endpoint rows. The implementations reduce that common core once, copy its compressed echelon form, and insert the few sector-specific endpoints.

The algebra behind the finisher is the following elementary lemma.

**Lemma 6.1** (Kernel-coordinate rank gain)**.** *Let $A_0$ be a row matrix over a field $K$, let $\{k_1, \dots, k_d\}$ be a basis of its right kernel, and let $B$ be a matrix of additional rows. Form the signature matrix*

$$\Sigma_{i,j} = \langle B_i, k_j \rangle. \tag{50}$$

*Then*

$$\operatorname{rank} \begin{pmatrix} A_0 \\ B \end{pmatrix} = \operatorname{rank} A_0 + \operatorname{rank} \Sigma. \tag{51}$$

*Proof.* The right kernel of the enlarged matrix is the kernel of the restriction $B|_{\ker A_0}$. In the chosen basis this restriction is $\Sigma$. Taking codimensions in the ambient column space gives (51). □

There are three instances of this decomposition in the computations.

(1) At prime level $N = P$, with $Q = NM$, $(N, M) = 1$, the $q \mid M$ rows form the sectorwise coprime core. By Theorem 4.2, the only necessary $N$-divisible top block is $q = N$. After complete-row replacement this block supplies the orbit space $W_N$ from (8). Its signatures on the direct sum of the sector kernels decide the final rank.

(2) At composite level $N = PN_0$, with $P$ occurring to exact exponent one in $N$ and $Q$, the $P$-free rows are again sectorwise. Every retained $P$-divisible row is evaluated on the direct sum of their kernels, and Lemma 6.1 gives its exact rank contribution.

(3) If an auxiliary prime $P_1$ divides $M$ to exact exponent one, the prime-level decomposition can be repeated inside each $N$-sector by writing

$$j = r + Nt, \qquad t = s + P_1 z.$$

Rows with $P_1 \nmid q$ preserve $(r, s)$; rows with $P_1 \mid q$ preserve $r$ and couple the $P_1$ values of $s$. This is the two-prime CRT implementation.

When $N$ and the split prime are odd and $Q$ is even, the CRT split combines with Section 2. In the prime-level coordinates $j = r + Nt$, the even-coordinate sector is characterized by

$$t \equiv r \pmod 2. \tag{52}$$

It therefore has $M/2$ columns for each $r$. Reflection supplies the other global parity, so it is never necessary to run both.

## 7. A rigorous sparse rank certificate

The largest common-core matrices are sparse and rectangular. Direct dense elimination is avoided by a block-Wiedemann calculation, but the raw output of one randomized run is not by itself a proof that the entire kernel was found. The rigorous certificate combines two inequalities obtained by different exact computations.

### 7.1. **Sparse square completion.** Let

$$C \in \mathbb{F}_2^{R \times D}, \qquad R \geq D,$$

be a tall common-core matrix. Choose $R - D$ distinct rows and let $G \in \mathbb{F}_2^{R \times (R-D)}$ have a unit column in each chosen row. Set

$$\mathcal{A} = [\, C \mid G \,] \in \mathbb{F}_2^{R \times R}. \tag{53}$$

Only $R - D$ nonzero entries are added. For every $G$, without a randomness hypothesis,

$$\ker \mathcal{A} \cap \{(x, y) : y = 0\} = \ker C \times \{0\}. \tag{54}$$

The completion can have a larger kernel than $C$; this is harmless once its full nullity is certified and the appended coordinates are eliminated exactly.

7.2. **The Berlekamp–Massey lower bound.** Choose block vectors $X, Y \in \mathbb{F}_2^{R\times b}$ and form the block-Wiedemann sequence

$$S_i = X^{\mathsf{T}} \mathcal{A}^{i+1} Y \in \mathbb{F}_2^{b\times b}. \tag{55}$$

The shift by one is essential. Each finite block Hankel matrix factors as

$$(S_{i+j})_{0\le i,j<t} = \big(X^{\mathsf{T}}\mathcal{A}^i\big)_{i=0}^{t-1} \mathcal{A} \big(\mathcal{A}^j Y\big)_{j=0}^{t-1}. \tag{56}$$

Hence every such Hankel rank is at most $\operatorname{rank} \mathcal{A}$. Exact block Berlekamp–Massey produces a minimal matrix generator $F(x) \in \mathbb{F}_2[x]^{b\times b}$ for this sequence. Its linear complexity is $\deg \det F$, so

$$d_{\mathrm{BM}} := \deg \det F \le \operatorname{rank} \mathcal{A}. \tag{57}$$

Unlucky random projections can make $d_{\mathrm{BM}}$ too small; they cannot make it too large.

The determinant degree is evaluated exactly without expanding the determinant. If $d_j$ is the degree of column $j$, define the reversed matrix

$$F^*(x) = F(x^{-1}) \operatorname{diag}(x^{d_1}, \dots, x^{d_b}). \tag{58}$$

Then

$$\deg \det F = \sum_{j=1}^{b} d_j - \operatorname{ord}_x \det F^*. \tag{59}$$

To find a valuation, compute the nullity of multiplication by $F$ on $(\mathbb{F}_2[x]/x^k)^b$. If $a_1, \dots, a_b$ are the local Smith exponents, this nullity is

$$\sum_{j=1}^{b} \min(a_j, k). \tag{60}$$

The first zero increment as $k$ increases proves that the valuation has stabilized. The matrices in this calculation have dimension $bk$, not $R$.

7.3. **The matching upper bound.** The Wiedemann reconstruction produces candidate vectors $z_1, \dots, z_k$. A separate exact pass checks

$$\mathcal{A} z_i = 0 \tag{61}$$

against every sparse row and row-reduces the candidates to prove their independence. Therefore

$$\operatorname{rank} \mathcal{A} \le R - k. \tag{62}$$

Combining (57) and (62) gives the central certificate:

$$\boxed{d_{\mathrm{BM}} = R - k} \quad \Longrightarrow \quad \operatorname{rank} \mathcal{A} = R - k, \quad \ker \mathcal{A} = \operatorname{span}\{z_1, \dots, z_k\}. \tag{63}$$

Several seeds may be used to obtain the $k$ independent vectors. Their randomness affects whether equality is reached, not the validity of (63) once reached.

Let $T\colon \ker \mathcal{A} \to \mathbb{F}_2^{R-D}$ retain the appended coordinates. If exact elimination gives $\operatorname{rank} T = t$, then

$$\operatorname{nullity} C = k - t, \qquad \operatorname{rank} C = D - k + t. \tag{64}$$

Indeed, (54) identifies $\ker C$ with $\ker T$. The resulting $k - t$ cleaned vectors are again checked against every literal row of $C$.

7.4. **Restoring the full row family.** The certified common-core kernel is only the first stage. The remaining steps are deterministic exact linear algebra:

(1) reflect the computed residue-parity core and insert the explicitly listed exceptional endpoint rows;
(2) insert every sector-specific endpoint from (49);
(3) translate the resulting sector kernels and form their direct sum;
(4) evaluate the retained coupling rows—the orbit rows at prime level, or all $P$-divisible rows at composite level—on that kernel;
(5) use Lemma 6.1 to compute the exact rank gain.

For a very wide final orbit-image matrix $B$ one may multiply on the right by an explicitly generated sparse hash matrix $P$ and reduce $BP$. If $B$ has $r$ rows and the computation finds

$$\operatorname{rank}(BP) = r, \tag{65}$$

then

$$r = \operatorname{rank}(BP) \le \operatorname{rank} B \le r,$$

so $B$ has full row rank. This is an exact compressed witness. A choice of $P$ for which (65) fails is merely inconclusive and is never reported as a rank defect.

Finally, Proposition 2.1 supplies the uncomputed global parity. If the resulting assisted matrix has full rank over $\mathbb{F}_2$, the nonzero binary maximal minor is odd as an integer, proving full rank over $\mathbb{Q}$ and $\mathbb{C}$.

**Theorem 7.1** (What constitutes a rigorous sparse certificate)**.** *The sparse pipeline proves the rank of the original integer row matrix when all of the following exact checks succeed:*

(1) *the exported sparse matrix agrees with the stated buffered rows;*
(2) *the block Berlekamp–Massey generator gives the lower bound* $d_{\mathrm{BM}}$ *in* (57)*;*
(3) $k$ *independent reconstructed vectors pass every row check in* (61)*;*
(4) *the equality* $d_{\mathrm{BM}} = R - k$ *in* (63) *holds;*
(5) *tail elimination, reflected exceptions, sector endpoints, and all coupling signatures are processed exactly;*
(6) *the final signature matrix, or an image as in* (65)*, has the required rank.*

*No assumption that a randomized block-Wiedemann run found the whole kernel is present in this conclusion. If items 2–4 do not meet exactly, the run has produced valid kernel vectors but no complete rank certificate.*

## 8. Implementation map

The programs below are included in `numerics/src/` in the accompanying repository. Their mathematical roles are as follows.

| Source | Role |
|---|---|
| `spatial_row_pruning.h` | Shared exact implementation of Theorem 4.3: saturated-anchor discovery, simultaneous proper-multiple deletion, and the reflection-compatible shift cap. |
| `spatial_qorder_f2_parity.c` | Direct dense $\mathbb{F}_2$ rank on one global parity; includes simultaneous triangular pruning. |
| `exact_q_rank_flint.c` | Exact characteristic-zero rank for Hall survivors. |
| `spatial_qorder_hall_screen_general_u64.c` | Production minimality pipeline: forced-divisor enumeration, rank upper bound, Hall flow, and in-process calls to the shared FLINT rank routine. |
| `spatial_crt_kernel_f2_parity_core.c` | Prime-level one-prime CRT core, sector endpoints, and orbit top-up. |
| `spatial_crt_kernel_f2_twoprime.c` | Second auxiliary-prime CRT split and exact projected finisher. |
| `spatial_crt_composite_f2_parity_parallel.c` | Composite-level large-prime-factor CRT with deterministic parallel signature reduction. |
| `spatial_crt_bw_export.c` | Export of one sparse parity common core. |
| `cado_square_complete_f2.c` | Sparse unit-column square completion (53). |
| `cado_rank_from_lingen_f2.c` | Exact computation of $\deg \det F$ from the block Berlekamp–Massey generator. |
| `cado_solution_check_f2.c` | Exact checks of reconstructed candidates against every square-matrix row, and their rank. |
| `spatial_crt_kernel_f2_bw_finish_minor.c` | Reflection, endpoints, kernel translations, and the compressed top-up witness. |

Supporting documentation and checks are included at these paths:

- `numerics/README.md`: build instructions, resource controls and certificate checks;
- `numerics/data/levels.tsv` and `numerics/data/minimality.tsv`: endpoint parameters and minimality targets;
- `numerics/n97/README.md`: the distributed sparse pipeline;
- `numerics/tests/smoke.py`: small regression checks;
- `numerics/tests/sparse_smoke.py`: the small sparse-pipeline check;
- `provenance/`: historical source hashes and validation records.

The small regression checks are not substitutes for the large production computations.